

User Guide

Introduction

SirDukeBox is a **desktop app** for managing instrument inventory, reservations, and finances.

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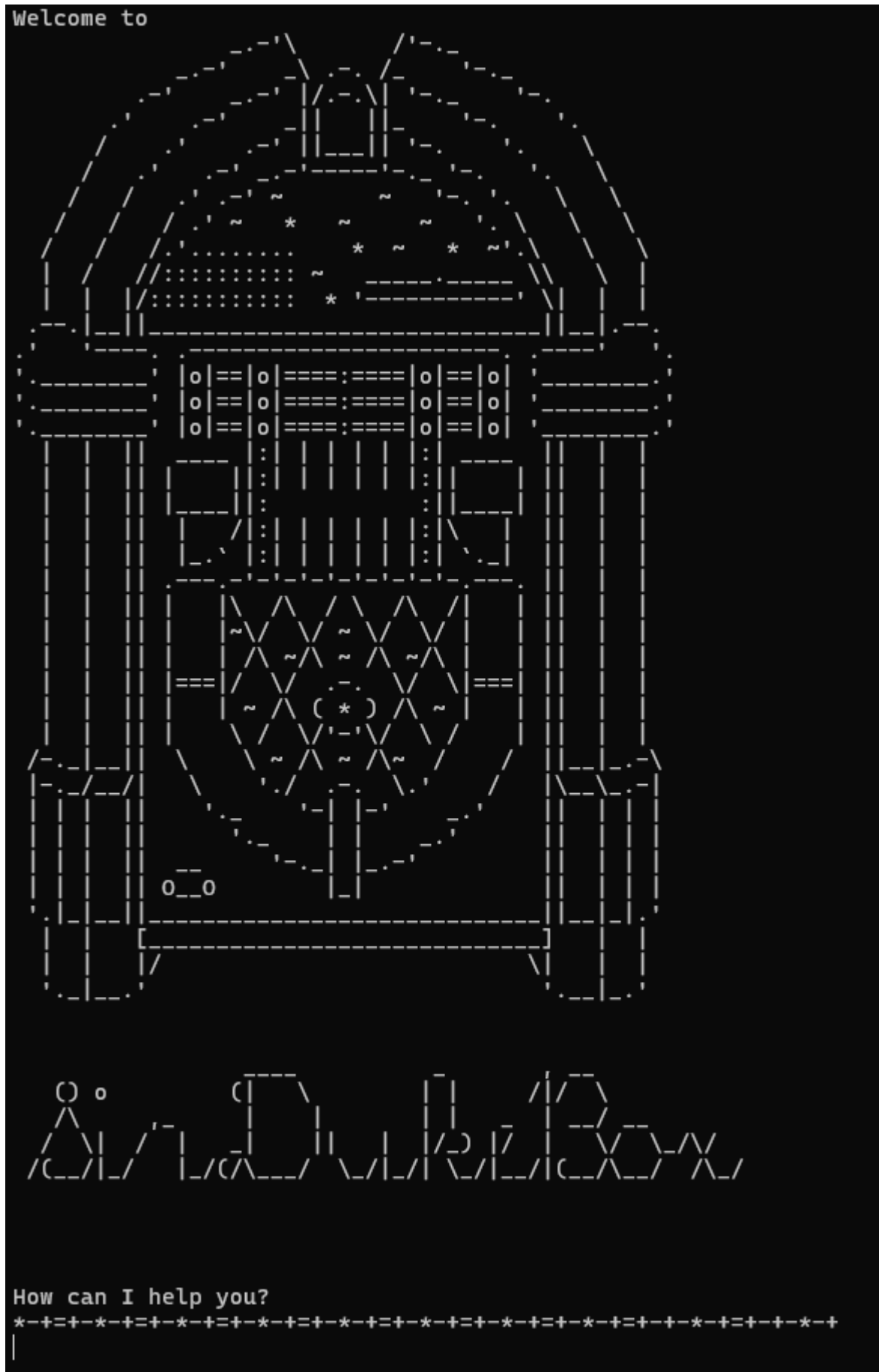
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Quick Start

1. Ensure that you have Java 17 or above installed.
2. Down the latest version of `SirDukeBox` from [here](#).
3. Copy the downloaded file to the folder you wish to use as the *home folder* for `SirDukeBox`
4. Open a command terminal, `cd` into the folder you put the jar file in, and use the `java -jar duke.jar` command to run the application.

5. Upon start up, `SirDukeBox` should reflect the following message in your terminal:



6. Type commands in the terminal and press Enter to execute. Here are some commands you could try:

- `help` : Lists all available commands for the user to input
- `list` : Lists all items currently in the database

- `exit` : Exits the app

7. Refer to **Features** below for a detailed list of all commands

Command Summary

Action	Format
List out Commands	<code>help</code>
Add Instrument	<code>add</code> <code>INSTRUMENT_NAME\ INSTRUMENT_MODEL\ INSTRUMENT_YEAR</code>
Delete Instrument	<code>delete INSTRUMENT_NUMBER</code>
List Instruments	<code>list</code>
List Instrument Stocks	<code>list stock</code>
Filtered List of Instruments	<code>list filter by: FILTER SEARCH_TERM</code>
List All <code>list</code> Features	<code>list help</code>
Reserve Instrument (without deadline)	<code>reserve INSTRUMENT_NUMBER</code>
Reserve Instrument (with deadline)	<code>reserve INSTRUMENT_NUMBER from: START_DATE to: END_DATE</code>
Extend Reservation	<code>extend INSTRUMENT_NUMBER to: END_DATE</code>
Return Instrument	<code>return INSTRUMENT_NUMBER</code>
User Commands Menu	<code>user</code>
Recommend Instrument	<code>recommend INSTRUMENT_NAME</code>
Finance Commands Menu	<code>finance</code>
Exit Program	<code>exit</code>

Notes about the command format:

- Words in `UPPER_CASE` are the parameters to be supplied by the user.
e.g. in `return INSTRUMENT_NUMBER`, `INSTRUMENT_NUMBER` is a parameter that can be used as `return 2`.
- `START_DATE` s and `END_DATE` s should be input in the format `dd/MM/yyyy`.
e.g. `extend 2 to: 07/08/2025` extends the deadline for reservation of instrument 2 to 7 August 2025.

- Instruments that have been reserved prior cannot be reserved again.
- Instruments that are not currently reserved cannot be returned.
- `INSTRUMENT_NUMBER` can be obtained by using `list` .
- By default, all newly added instruments are not reserved.

Features

Instrument Commands

Listing Features

`SirDukeBox` provides several listing functionalities to help navigate the `SirDukeBox` inventory more easily.

List Instruments - `list`

Lists instruments stored in the database according to their INSTRUMENT_NUMBER

Format: list

Example of usage:

list

Expected output:

$$*-+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ -*+=+ *$$

Here is the list of instruments:

1. Guitar | Yamaha | 1989 | Rented | Rented from: 1/4/2025 | Rented to: 2/4/2025

2. Piano | Yamaha | 2016

3. Flute | Yamaha | 2023

4. Guitar | Fender | 1962

5. Guitar	Fender	2020
-----------	--------	------

6. Piano	Bosendorfer	2023	Rented
----------	-------------	------	--------

7. Piano | Steinway & Sons | 2020

-+=+--+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+-----

List of Instrument Stocks - `list stock`

Lists the total, rented and available quantity for all instruments in the database according to their INSTRUMENT_NAME

Format: `list stock`

Example of usage:

```
list stock
```

Expected output:

[illegible]

Filtered List of Instruments - list filter by:

Lists the relevant instruments according to the `FILTER` and `SEARCH_TERM` provided by the user.

Format: list filter by: FILTER SEARCH_TERM

- Possible combinations of `FILTER` and `SEARCH_TERM`
 - `FILTER : name , SEARCH_TERM : INSTRUMENT_NAME`
 - Filters instruments according to the specified `INSTRUMENT_NAME`
 - `FILTER : model , SEARCH_TERM : INSTRUMENT_MODEL`
 - Filters instruments according to the specified `INSTRUMENT_MODEL`
 - `FILTER : year , SEARCH_TERM : INSTRUMENT_YEAR`
 - Filters instruments according to the specified `INSTRUMENT_YEAR`
 - `FILTER : reserved ( SEARCH_TERM is not required)`
 - Filters instruments that `is_rented()`
 - `FILTER : available ( SEARCH_TERM is not required)`
 - Filters instruments that are not `is_rented()`
 - `FILTER : overdue ( SEARCH_TERM is not required)`
 - Filters instruments that `is_overdue()`

Example of usage:

```
list filter by: model Yamaha
```

Expected output:

```
*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=
Here is the list of instruments:
1. Guitar | Yamaha | 1989 | Rented | Rented from: 1/4/2025 | Rented to: 2/4/2025
2. Piano | Yamaha | 2016
3. Flute | Yamaha | 2023
*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=
```

List all `list` functionalities - `list help`

Lists all available `list` functionalities. A brief description for each command is included.

Format: `list help`

Expected output:

```
list help
*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+
Here is a list of available subcommands:
Available Subcommands:

help: list all subcommands for `list`
stock: list total, available and reserved quantities for each instrument
filter by <FILTER> <SEARCH_TERM>: list instruments according to FILTER and SEARCH_

Available FILTERs and SEARCHTERMs:

FILTER: name, SEARCH_TERM: INSTRUMENT_NAME
FILTER: model, SEARCH_TERM: INSTRUMENT_MODEL
FILTER: year, SEARCH_TERM: INSTRUMENT_YEAR
FILTER: reserved (SEARCH_TERM is not required)
FILTER: available (SEARCH_TERM is not required)
*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+
```

Adding an Instrument - `add`

Adds a new instrument to the database.

Format: `add INSTRUMENT_NAME | INSTRUMENT_MODEL | INSTRUMENT_YEAR`

- `INSTRUMENT_NAME` refers to the type of instrument being added (e.g. "Guitar")
- `INSTRUMENT_MODEL` refers to the brand/model of the specific instrument being added (e.g. "Yamaha")
- `INSTRUMENT_YEAR` refers to the year of production of the instrument being added (e.g. "1989")
 - lowest value for `INSTRUMENT_YEAR` is set to 1600 - any year before that is invalid

Example of usage:

```
add Guitar|Yamaha|1989
```

Expected output:

```
*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+*+=+
Here is the list of instruments:
1. Guitar | Yamaha | 1989
```

```
*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=*-+=
Would you like to assign this instrument to a user? [Y/N]
```

Adding an instrument allows you to assign it to a user

Deleting an Instrument - delete

Deletes an existing instrument from the database.

Format: delete INSTRUMENT_NUMBER

- `INSTRUMENT_NUMBER` refers to the number assigned to the particular instrument. Use `list` to view each instrument's `INSTRUMENT_NUMBER`

Example of usage:

```
delete 1
```

Expected output:

```
Removed instrument [Guitar | Yamaha | 1989 | Assigned to: Kash] from user [Kash]
Deleting instrument: Guitar | Yamaha | 1989 | Assigned to: Kash
Now you have 0 instruments
```

-+=+-+=+*-+=+*-+=+*-+=+*-+=+*-+=+*-+=+*-+=+*-+=+

Reserving an Instrument (without deadline) - reserve

Reserves an unreserved instrument in the database.

Format: reserve INSTRUMENT_NUMBER

- `INSTRUMENT_NUMBER` refers to the number assigned to the particular instrument. Use `list` to view each instrument's `INSTRUMENT_NUMBER`
- `SirDukeBox` will set the `START_DATE` as today's date *
 - `START_DATE` refers to the date at which the reservation commences

Example of usage:

reserve 6 (suppose today's date is 07/04/2025)

Expected output:

Reserving instrument: Piano | Bosendorfer | 2023

[illegible]

Here is the list of instruments:

- | | | | | | |
|-----------|--------|------|--------|-----------------------|---------------------|
| 1. Guitar | Yamaha | 1989 | Rented | Rented from: 1/4/2025 | Rented to: 2/4/2025 |
| 2. Piano | Yamaha | 2016 | | | |
| 3. Flute | Yamaha | 2023 | | | |

- [illegible]

Reserves an unreserved instrument in the database from a specified `START_DATE` to a specified `END_DATE` .

- `INSTRUMENT_NUMBER` refers to the number assigned to the particular instrument. Use `list` to view each instrument's `INSTRUMENT_NUMBER`
- `START_DATE` refers to the date at which the reservation commences
- `END_DATE` refers to the date at which the reservation ends
- Both `START_DATE` and `END_DATE` are in the format: `(dd/mm/yyyy)`

Expected output:

-+=+--+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-*-+=+-+*

1.	Guitar		Yamaha		1989		Rented		Rented from:	1/4/2025		Rented to:	2/4/2025
2.	Piano		Yamaha		2016								
3.	Flute		Yamaha		2023								
4.	Guitar		Fender		1962								
5.	Guitar		Fender		2020		Rented		Rented from:	1/4/2025		Rented to:	2/4/2025
6.	Piano		Bosendorfer		2023		Rented		Rented from:	7/4/2025			
7.	Piano		Steinway & Sons		2020								

-+ = +--+ = +-*-+ = +-*-+ = +-*-+ = +-*-+ = +-*-+ = +-*-+ = +-*-+ = +-*-+

Extends the reservation period of a reserved instrument

- `INSTRUMENT_NUMBER` refers to the number assigned to the particular instrument. Use `list` to view each instrument's `INSTRUMENT_NUMBER`
- `END_DATE` refers to the date at which the reservation ends
- `END_DATE` is in the format: (dd/mm/yyyy)

Example of usage:

extend 5 to: 04/04/2025

Expected output:

Extending reservation of instrument: Guitar from 2025-04-01 to 2025-04-04

[illegible]

Here is the list of instruments:

- | | | | | | | | | | | | |
|----|--------|--|-----------------|--|------|--|--------|--|-----------------------|--|---------------------|
| 1. | Guitar | | Yamaha | | 1989 | | Rented | | Rented from: 1/4/2025 | | Rented to: 2/4/2025 |
| 2. | Piano | | Yamaha | | 2016 | | | | | | |
| 3. | Flute | | Yamaha | | 2023 | | | | | | |
| 4. | Guitar | | Fender | | 1962 | | | | | | |
| 5. | Guitar | | Fender | | 2020 | | Rented | | Rented from: 1/4/2025 | | Rented to: 4/4/2025 |
| 6. | Piano | | Bosendorfer | | 2023 | | Rented | | | | |
| 7. | Piano | | Steinway & Sons | | 2020 | | | | | | |

[illegible]

Returning an Instrument - return

Returns a reserved instrument

Format: return INSTRUMENT_NUMBER

- `INSTRUMENT_NUMBER` refers to the number assigned to the particular instrument. Use `list` to view each instrument's `INSTRUMENT_NUMBER`

Example of usage:

```
return 5
```

Expected output:

Returning instrument: Guitar | Fender | 2020 | Rented | Rented from: 1/4/2025 | Re

[illegible]

Here is the list of instruments:

1. Guitar | Yamaha | 1989 | Rented | Rented from: 1/4/2025 | Rented to: 2/4/2025
2. Piano | Yamaha | 2016
3. Flute | Yamaha | 2023
4. Guitar | Fender | 1962
5. Guitar | Fender | 2020
6. Piano | Bosendorfer | 2023 | Rented
7. Piano | Steinway & Sons | 2020

[illegible]

User commands - user

User will bring up a list of commands that pertains to users:

1. Adding a user
2. Removing a user
3. Listing all users
4. Listing a user's instruments ...which is selected by inputting the index of the command and following onscreen instructions

For option 4 , the user will be able to choose from 2 lists:

1. Rental History
 - This is the list of all instruments that have been added to the user during the current session (including deleted instruments)
2. Currently Instruments
 - This is the list of instruments still allocated to the user (modified after deleting)

For example, if the current list of instruments assigned to the user is:

- ```
1. Guitar | Yamaha | 2000
2. Piano | Yamaha | 2000
```

...and delete 2 is inputted, CurrentInstruments will be:

1. Guitar | Yamaha | 2000

...while RentalHistory will be:

- ```
1. Guitar | Yamaha | 2000
2. Piano | Yamaha | 2000
```

Example of usage:

Input:

user

Expected output:

[illegible]

Input:

4

Expected output:

[illegible]

Input:

1

Expected output:

```
*+*=+*-+*=+*-+*=+*-+*=+*-+*=+*-+*=+*-+*=+*-+*-+
Available List Choices:
1. Rental History
2. Current Instruments
```

Input:

1

Expected output:

```
*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+
Here is the list of instruments:
1. Piano | Yamaha | 2000 | Assigned to: Teng Wai
*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+-+-*-+
```

...or if list is empty:

[illegible]

Recommendation for an instrument - recommend

Recommends an instrument from existing instrument list/inventory that is popular and not rented

Format: recommend INSTRUMENT_NAME

- The `INSTRUMENT_NAME` can only be "Guitar", "Piano" or "Flute"

Example of usage:

recommend Guitar

Expected output:

```
*+#+*-+#+*-+#+*-+#+*-+#+*-+#+*-+#+*-+#+*+
Here is our recommendation:
```

```
5. Guitar | Fender | 2020
```

```
*+#+*-+#+*-+#+*-+#+*-+#+*-+#+*-+#+*-+#+*+
```

Finance commands - `finance`

Finance commands pertain to managing the finances related to instruments

1. List all finance commands `help`
2. Track incoming payment `add: AMOUNT`
3. Track outgoing payment `subtract: AMOUNT`
4. Get current balance `get`

...which is selected by inputting the index of the command and following onscreen instructions

When reserving items balance will increase, when returning overdue items the application will automatically calculate and add how much the user owes you.

Example of usage: Input:

```
finance add: 50
```

Expected output:

[illegible]

List out Commands - help

Lists all available commands. A brief description for each command is included.

Format: `help`

Example of usage:

```
help
```

Expected output:

```
Available Commands:
help: list all commands
list: list all instruments
add: adds a new instrument
delete: deletes an existing instrument
reserve: reserves an available instrument
extend: changes the return date of a reserved instrument
return: returns a reserved instrument
user: choose user commands
recommend: recommends a recommended instrument
finance: Manage finances: (use help flag to see commands)
exit: quit SirDukeBox
```

Exiting the Program - `exit`

Exits the program

Format: `exit`

Saving the data

SirDukeBox data are saved automatically as .txt files in the same directory as the `sirDukeBox.jar` file.

The files are saved as `SirDukeBox.txt` and `DukeFinance.txt`

Loading the data

Existing SirDukeBox data (from a previous session) are loaded automatically upon start up

NOTE: Any changes made to the .txt files outside of the functionalities of SirDukeBox (e.g. editing the .txt file manually in notepad) may render its format invalid. This may result in SirDukeBox being unable to load an existing .txt file

FAQ

How do I transfer my data to another computer?

Having installed the app in the other computer, overwrite the empty data files created (`SirDukeBox.txt` and `DukeFinance.txt`) with the existing files that contains the data from your original computer.

Should I edit the save files manually?

SirDukeBox save files are not meant to be edited manually. Editing the save files manually in a text editor runs the risk of corrupting the existing data, after which SirDukeBox may be unable to load the file upon start up.

Can I add instruments of the same `INSTRUMENT_NAME` , `INSTRUMENT_MODEL` and `INSTRUMENT_YEAR` ?

Yes, in the event that you have two identical instruments, SirDukeBox allows duplicate entries of instruments with the same `INSTRUMENT_NAME` , `INSTRUMENT_MODEL` and `INSTRUMENT_YEAR` .

Can I return a reserved instrument before its `START_DATE` ?

No, SirDukeBox requires that instruments must be rented for at least one day from its `START-DATE` . If the `START_DATE` has yet to pass, it is assumed that the instrument has not been loaned out for a reservation yet.